

PRACTICAL 8

Static routing

Static routing is a networking technique where a network administrator manually configures fixed paths in a router's routing table, rather than relying on automatic, dynamic, or adaptive protocols. It is commonly used for smaller networks or to define default routes, providing high security and low CPU overhead for routers, but requiring manual updates if the network topology changes.

Routing information protocol(RIP)

Routing information protocol(RIP) is a dynamic, distance-vector routing protocol used by routers to determine the best path for data packets across small-to-medium networks. It selects the shortest path based on hop count (maximum 15), broadcasting its entire routing table to neighbors every 30 seconds.

Difference

Static routing involves manually configuring routes by an administrator, making it ideal for small, secure, and stable networks, whereas RIP is a dynamic protocol that automatically learns and updates routes, making it better for larger, changing networks

Advantages

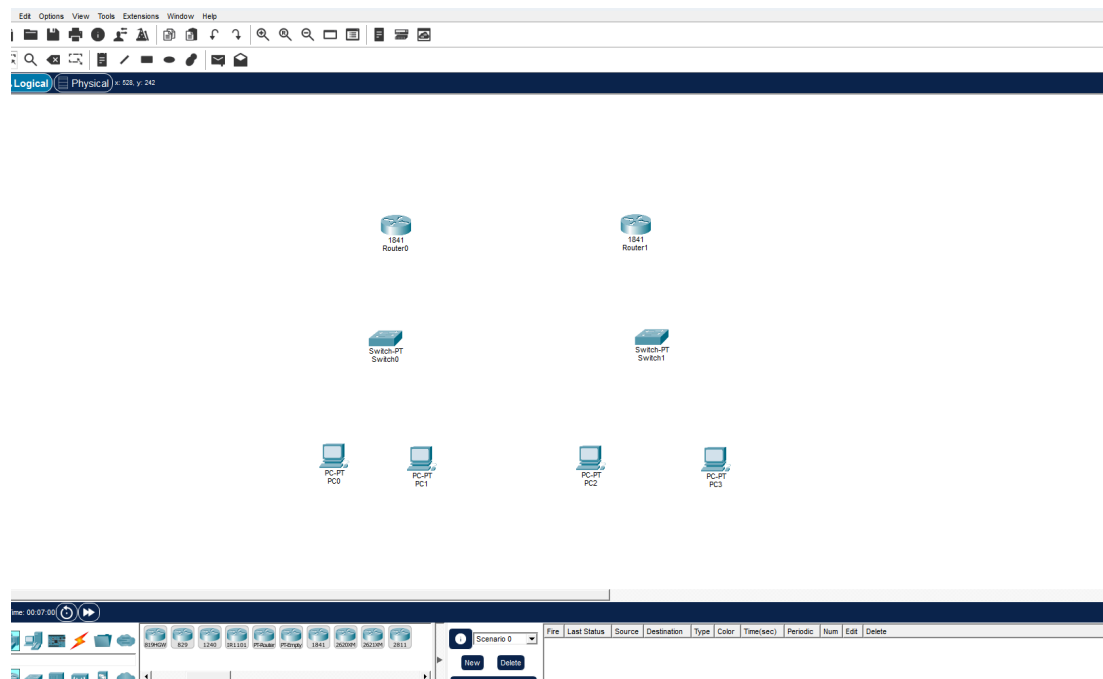
Static routing requires manual configuration of paths by an administrator, making it ideal for small, stable networks, while **RIP** is a dynamic, distance-vector protocol that automatically shares routing information, suitable for smaller, less complex environments

Disadvantages

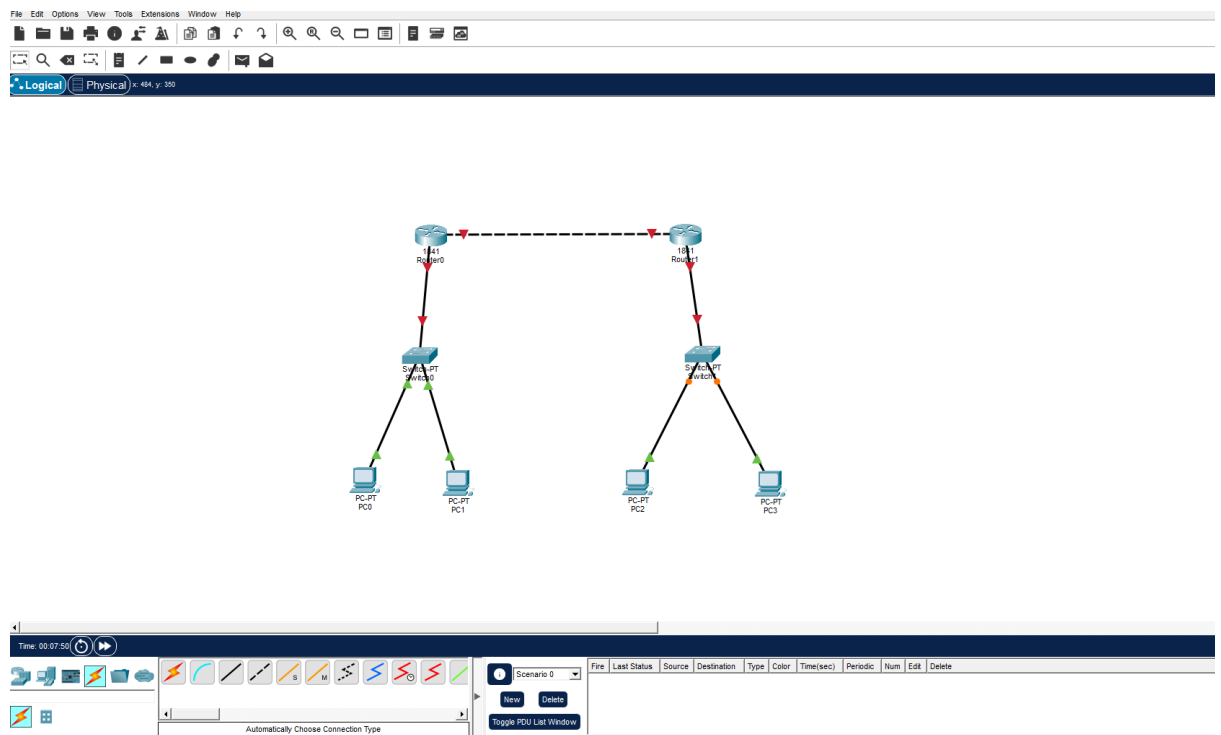
Static routing highly impractical for large networks as it becomes difficult to manage, configure, and maintain and RIP also has a maximum hop count of 15; a destination 16 hops away is considered unreachable, making it unsuitable for large networks.

Procedure

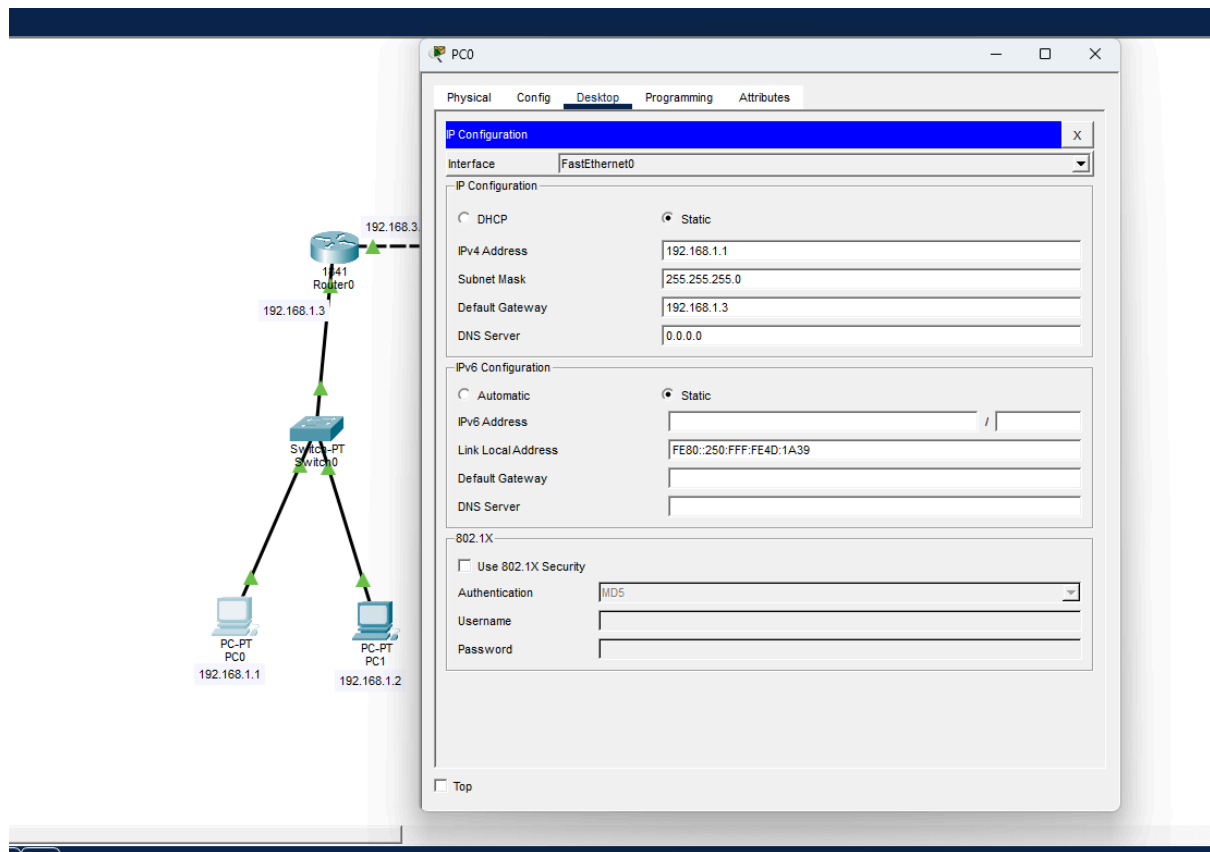
1) add all devices i.e 2 routers(1841), 2 switches(PT-Switch), and 4 devices(PC)



2) make connection by cables

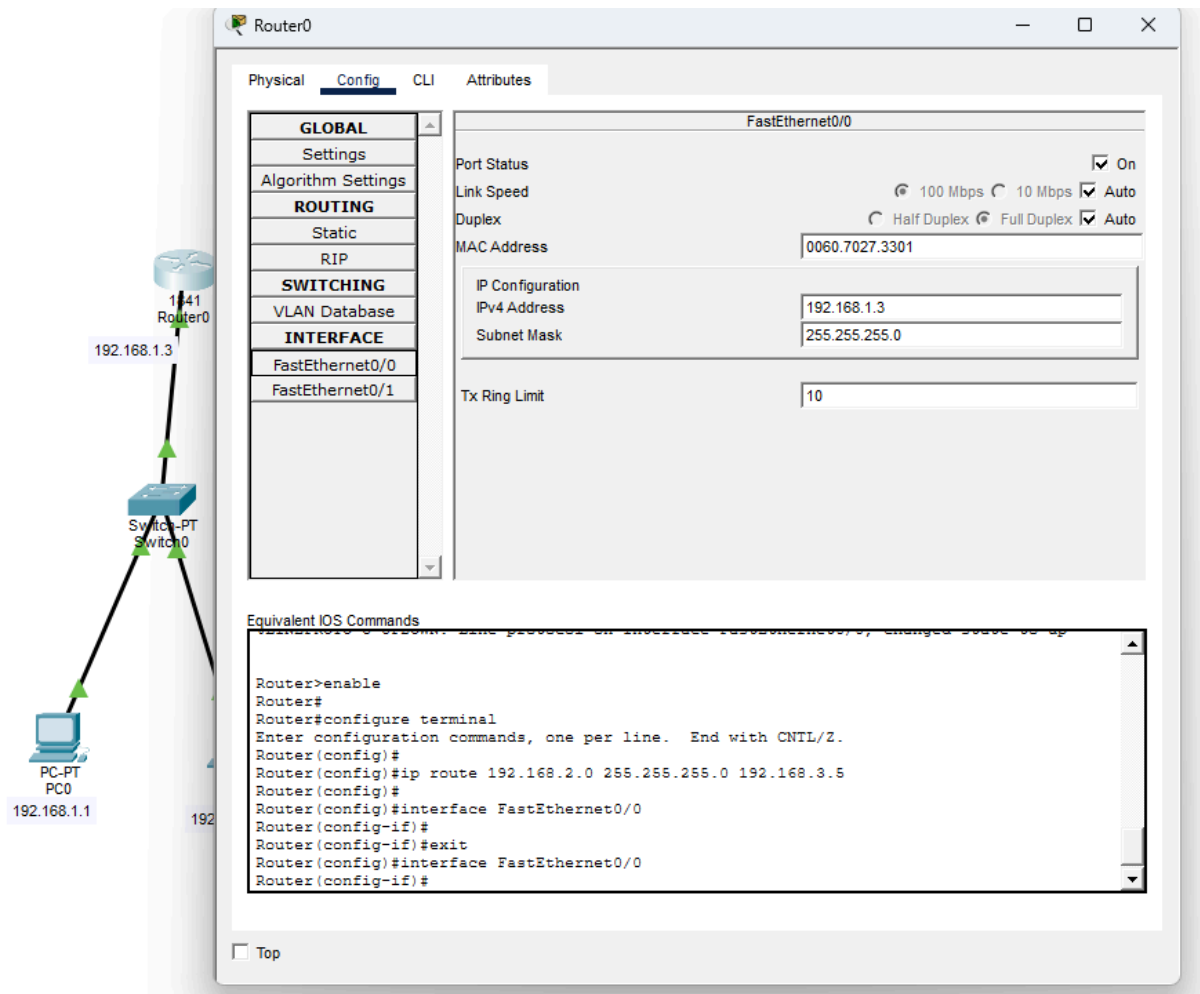


3) Give IP address to all the devices



4) Now check the cable name of cable connecting switch 0 and router 0 and also router 0 and router1 .then click on router 1 ->config -> interface ->fast ethernet0/0 and add ip address to cable with address of first network to which it is connected make sure that port status is on.

In fast Ethernet0/1 give a address other than the two networks



The image shows a network diagram on the left and the configuration window for Router0 on the right.

Network Diagram:

- A PC-PT (PC0) with IP 192.168.1.1 is connected to a Switch-PT (Switch0).
- Switch0 is connected to Router0 (192.168.1.3).
- Router0 is connected to another Router (192.168.1.4).

Router0 Configuration Window:

The window is titled "Router0" and has tabs for Physical, Config, CLI, and Attributes. The "Config" tab is active, showing a tree view on the left with categories: GLOBAL, ROUTING, SWITCHING, and INTERFACE. The "INTERFACE" category is expanded, showing "FastEthernet0/0" and "FastEthernet0/1".

The configuration for "FastEthernet0/0" is shown on the right:

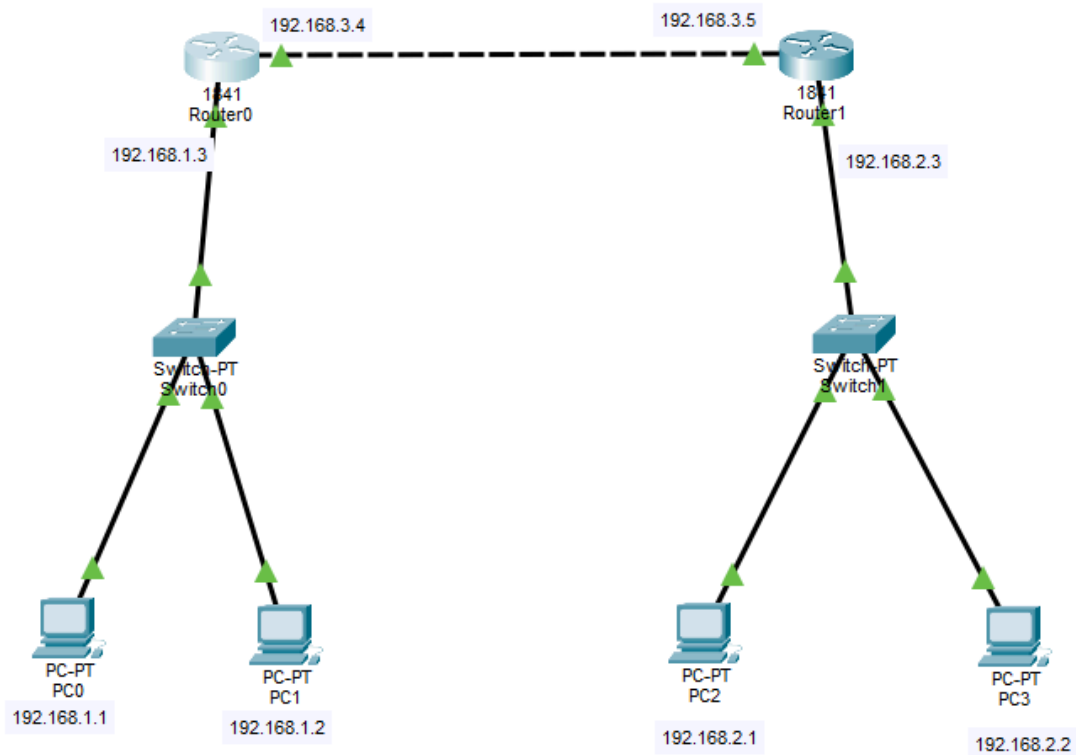
- Port Status: ☒ On
- Link Speed: ☒ 100 Mbps ☐ 10 Mbps ☒ Auto
- Duplex: ☐ Half Duplex ☒ Full Duplex ☒ Auto
- MAC Address: 0060.7027.3301
- IP Configuration:
 - IPv4 Address: 192.168.1.3
 - Subnet Mask: 255.255.255.0
- Tx Ring Limit: 10

Below the configuration fields, there is a section for "Equivalent IOS Commands" with a text area containing the following commands:

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#ip route 192.168.2.0 255.255.255.0 192.168.3.5
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
```

At the bottom left of the window, there is a "Top" button.

5) now in router 1 add the address of both cables as done in router 0 i.e 0/0 and 0/1.now all address are assigned



6) add gateway to all pc's and fill ip address of router they are connector to in gateway

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

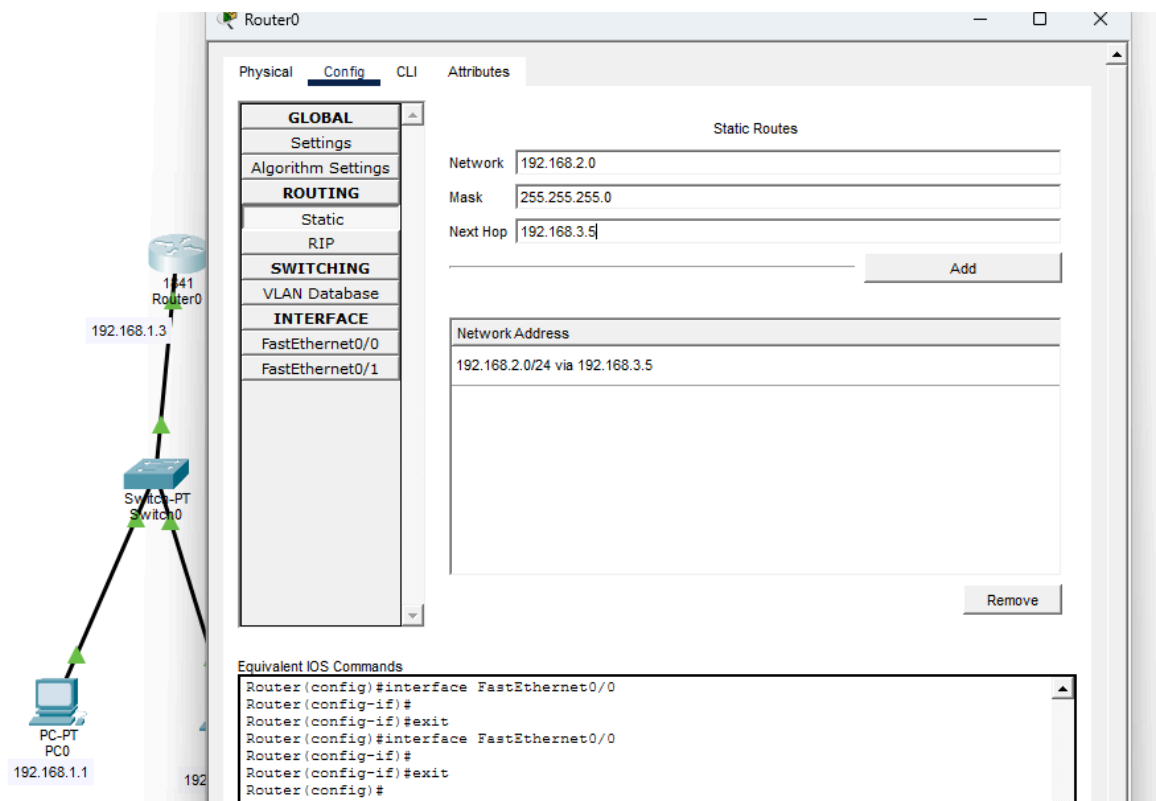
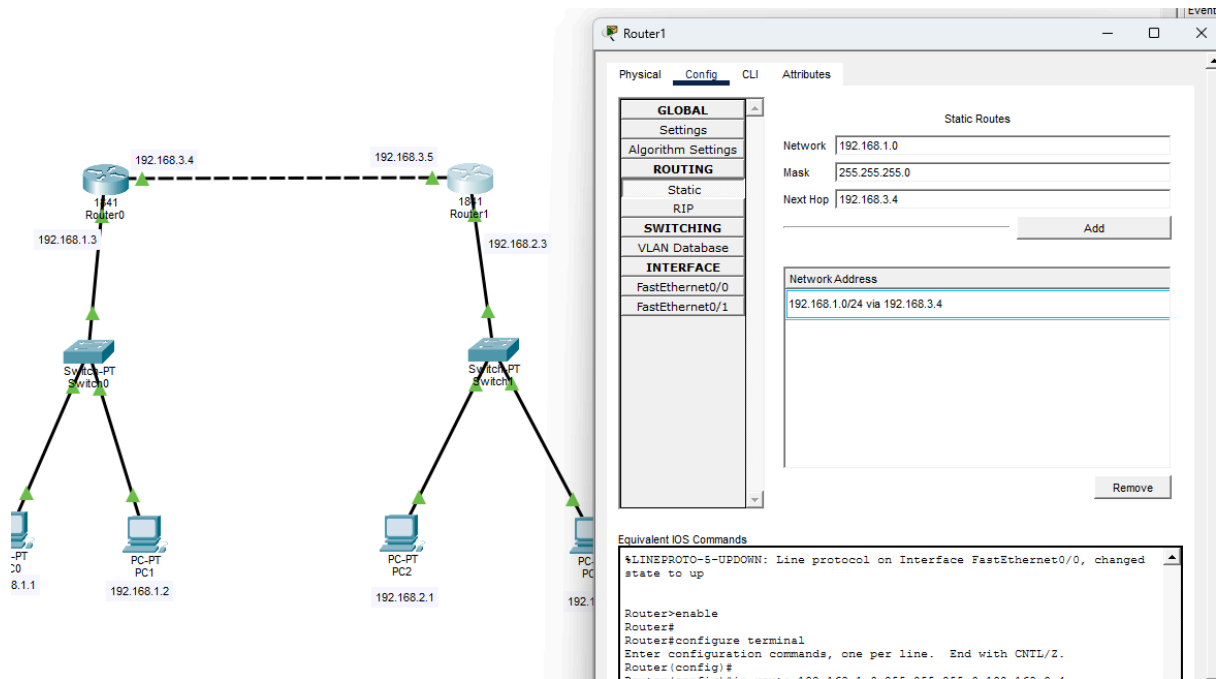
IPv4 Address 192.168.1.1

Subnet Mask 255.255.255.0

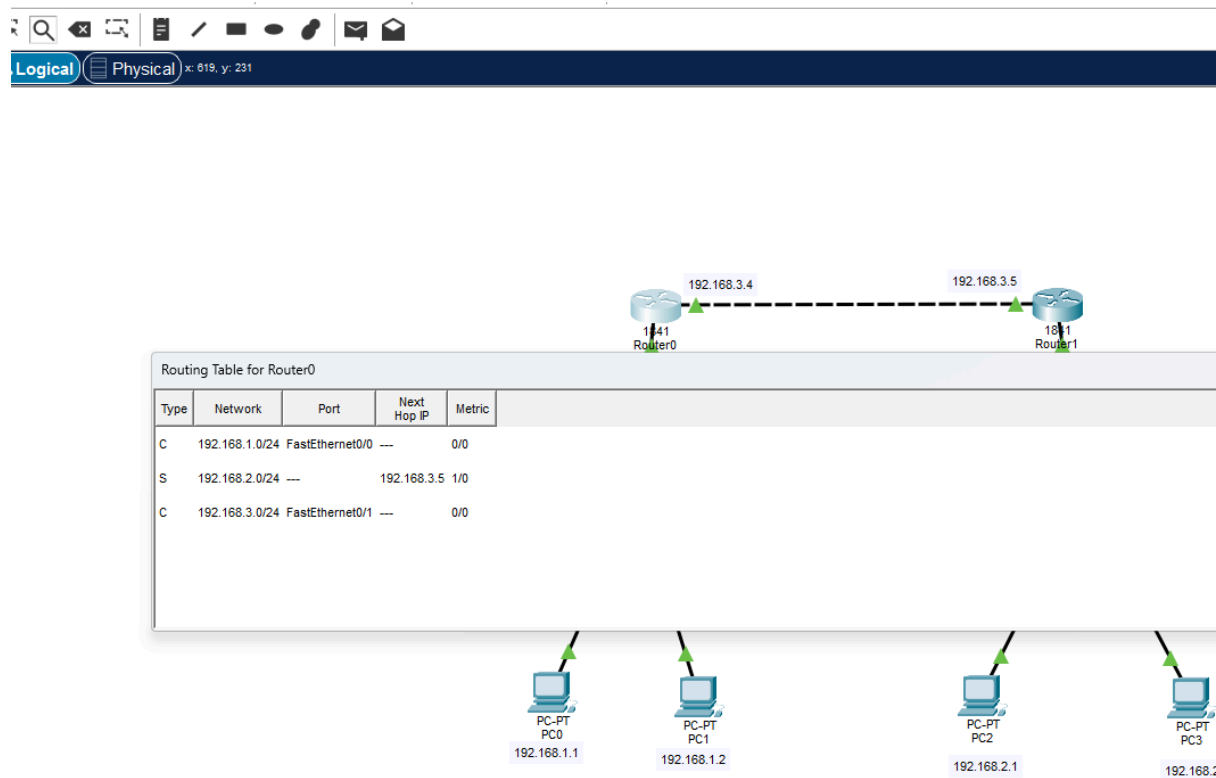
Default Gateway 192.168.1.3

DNS Server 0.0.0.0

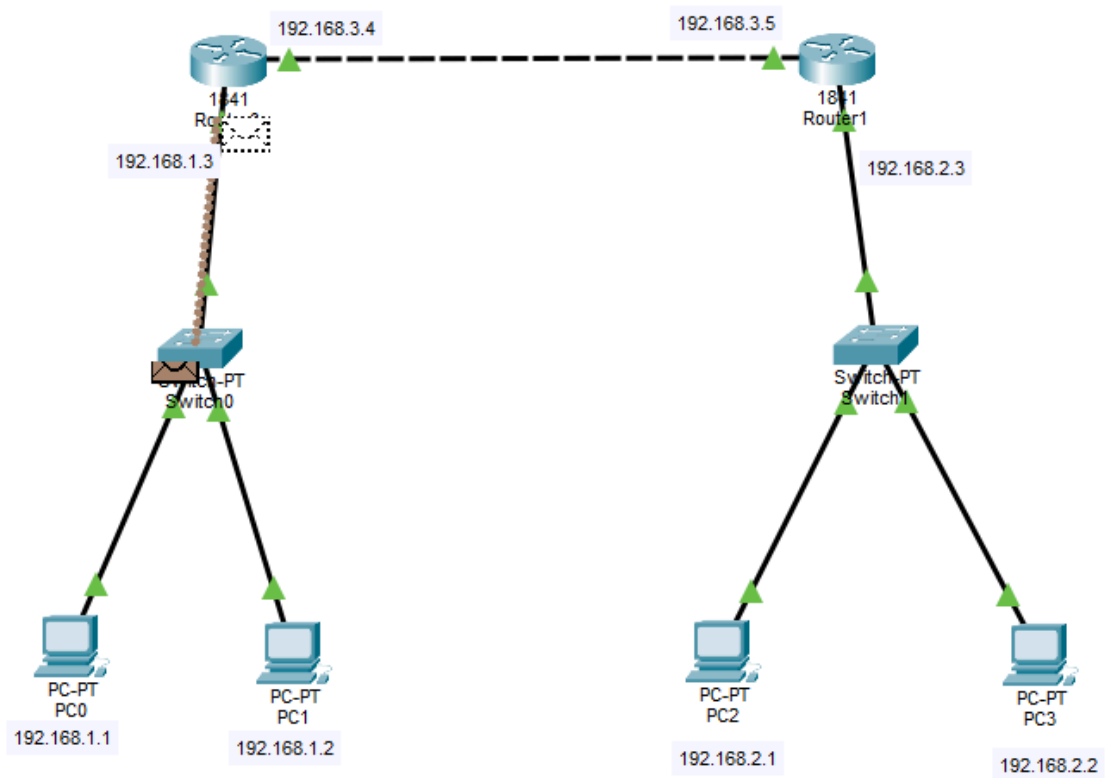
7) now connect both routers by config-> Static and setting connection between both routers



8) to check the connections of the router click on search icon ->router whose connections to be checked ->router table



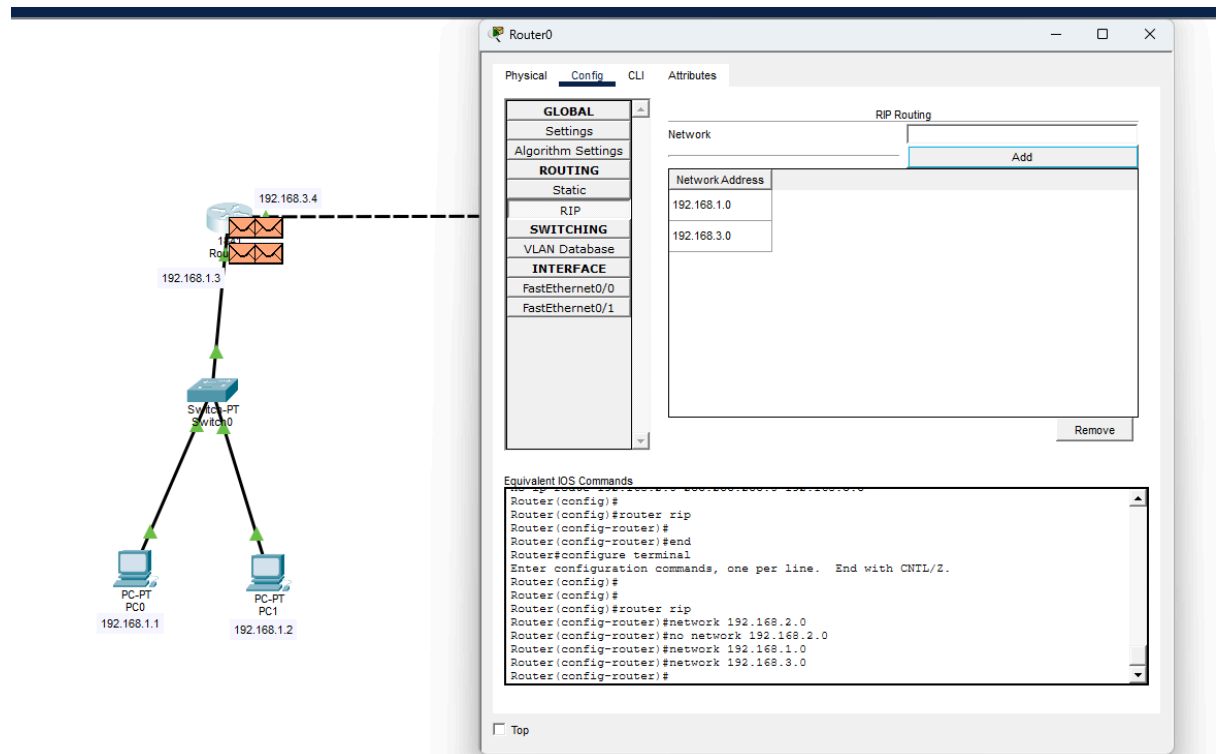
9) now pass the message and check if connection is successfully established



RIP ROUTING

PROCEDURE

- 1) Same as static only difference is in step 7 i.e connect both routers by config-> RIP and setting connection between both routers



The image displays a network topology and the configuration interface for Router0.

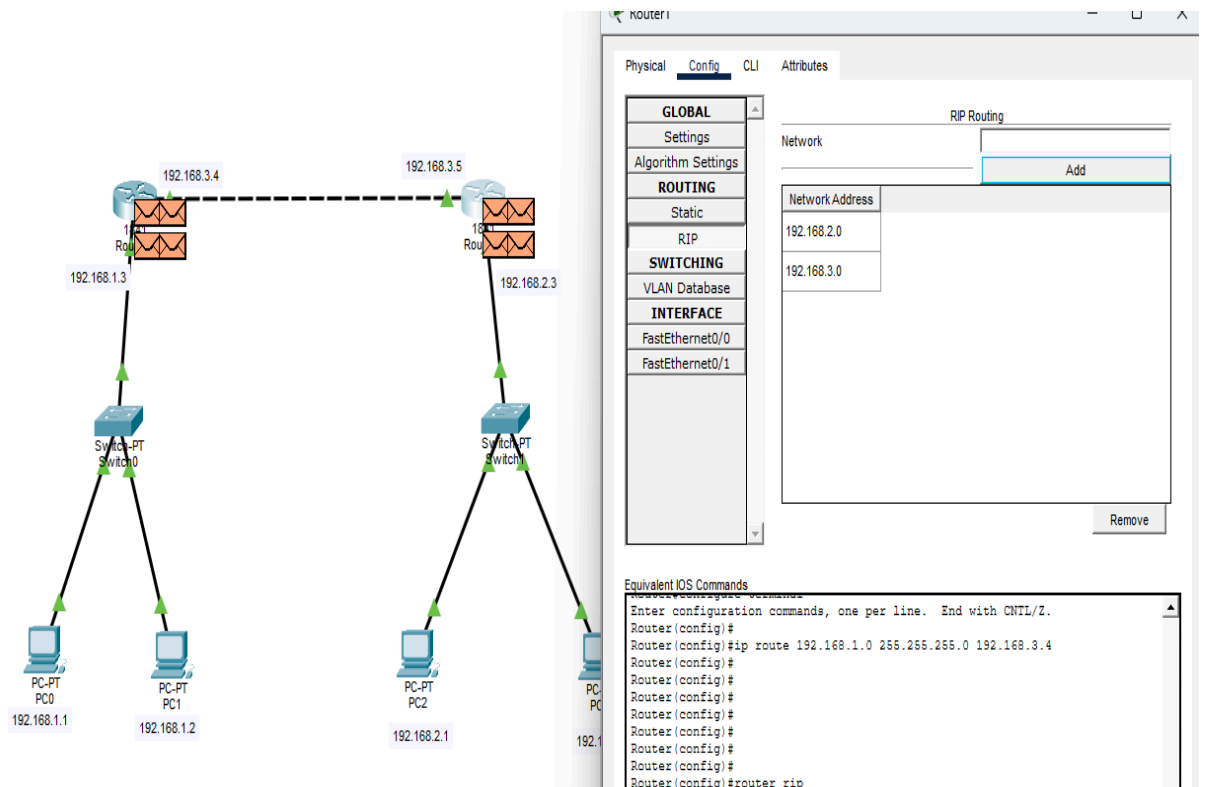
Network Diagram: A central Switch-PT (Switch0) is connected to two PCs (PC0 and PC1). PC0 has IP 192.168.1.1 and PC1 has IP 192.168.1.2. The switch is connected to a Router0, which has IP 192.168.1.3 on its interface connected to the switch. Router0 is also connected to another router (192.168.3.4) via a dashed line representing a connection.

Router0 Configuration Window: The window shows the configuration for Router0. The left sidebar lists various configuration sections: GLOBAL, Settings, Algorithm Settings, ROUTING, Static, RIP, SWITCHING, VLAN Database, INTERFACE, FastEthernet0/0, and FastEthernet0/1. The main area is titled "RIP Routing" and contains a table for "Network Address" with the following entries:

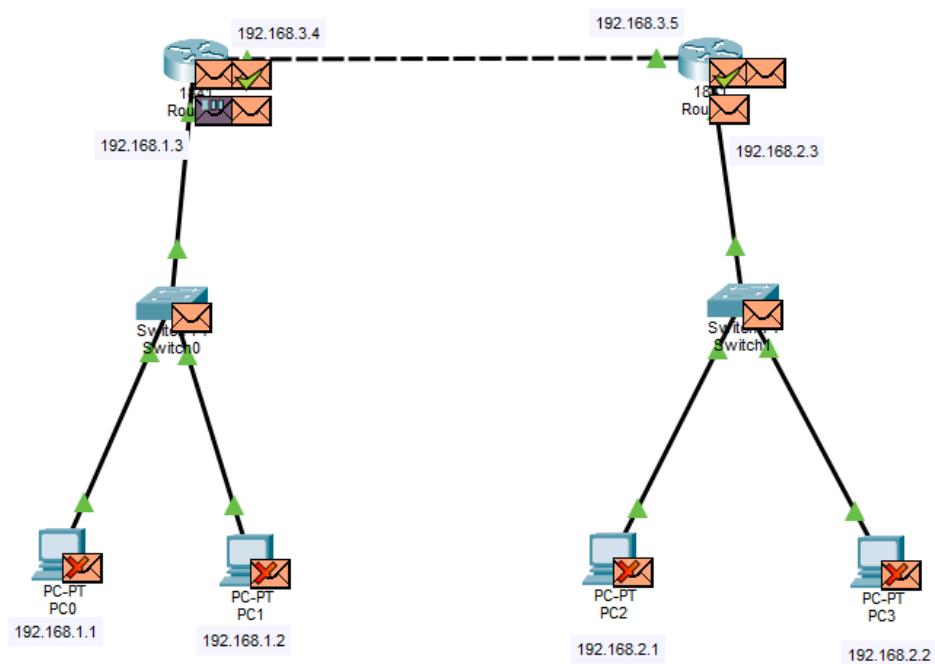
Network Address
192.168.1.0
192.168.3.0

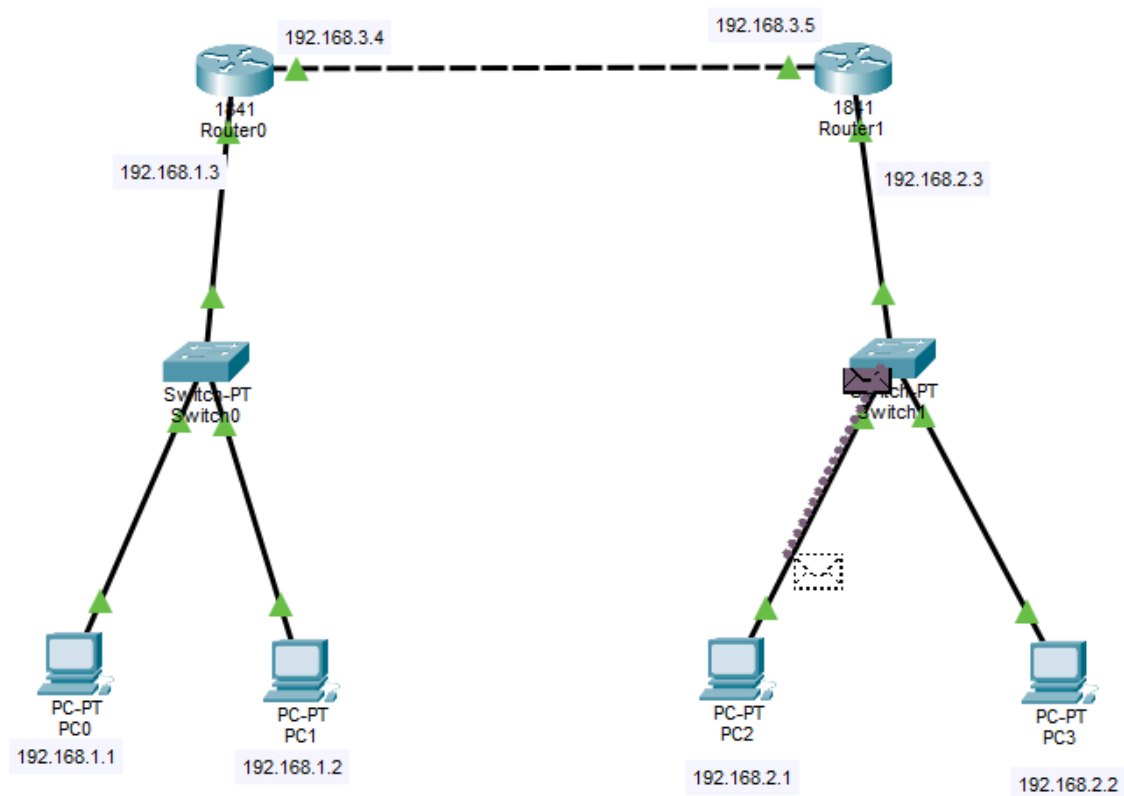
Below the table, there is a "Remove" button. At the bottom, the "Equivalent IOS Commands" section shows the following configuration commands:

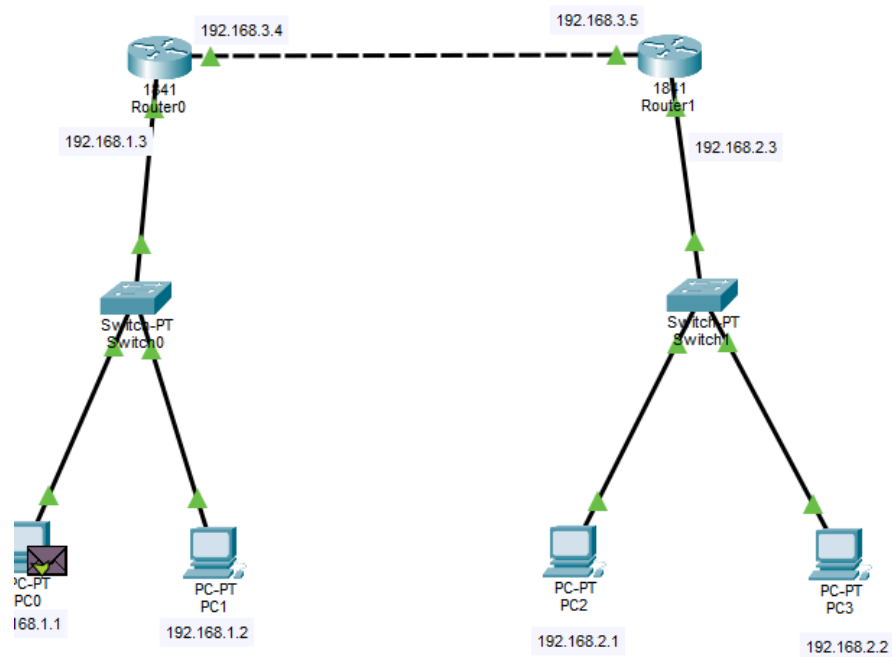
```
Router(config)#  
Router(config)#router rip  
Router(config-router)#  
Router(config-router)#end  
Router#configure terminal  
Enter configuration commands, one per line. End with CNTL/Z.  
Router(config)#  
Router(config)#router rip  
Router(config-router)#network 192.168.1.0  
Router(config-router)#no network 192.168.2.0  
Router(config-router)#network 192.168.1.0  
Router(config-router)#network 192.168.3.0  
Router(config-router)#
```



2) MESSAGE IS SUCCESSFULLY SENT







Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Del
...	Successful	PC0	PC2	ICMP		0.000	N	0	(e...	